

	Since the ratio k/m is not a constant, the student's suggestion is incorrect.
2(a)(i)	Both trolleys could not be stationary at the same time as this would imply that the total momentum of the system is zero at that instant. Since the total momentum of the system before collision was $(1.5)(0.90) + (1.2)(-2.2) = -1.29 \text{ kg m s}^{-1}$, with no external resultant force acting on the system, the total momentum should remain constant at this value throughout, according to the Principle of Conservation of Momentum.
2(a)(ii)	According to Newton's third law, the forces on trolleys A and B form an action-reaction pair. Hence, the average force between the trolleys may be determined by calculating the average force on trolley A. According to Newton's second law of motion, the average force on A = (change in momentum of A) / time $= [(1.5)(-0.70) - (1.5)(0.90)]/0.30$ $= -8.0 \text{ N}$
2(b)(i)	Let the speed of trolley B after the collision be v. By Newton's third law, the average force on trolley B is equal in magnitude but opposite in direction to the average force on trolley A. Hence, the average force on trolley B = 8.0 N. $\Rightarrow$ (change in momentum of B) / time = 8.0 N $\Rightarrow [(1.2)(v) - (1.2)(-2.2)]/0.30 = 8.0$ $\Rightarrow v = -0.2 \text{ m s}^{-1}$ Therefore, final speed of trolley B after collision is 0.2 m s^{-1}.
2(b)(ii)	Since the rightward direction was assigned as the positive direction, the negative sign in (b)(i) implies that trolley B was still moving towards the left after the collision, i.e. still in the same direction as that of its initial velocity.
2(c)	Relative speed of approach = $0.90 - (-2.2) = 3.1 \text{ m s}^{-1}$ Relative speed of separation = $-0.2 - (-0.70) = 0.50 \text{ m s}^{-1}$ Since the relative speed of approach is not equal to the relative speed of separation, the collision is inelastic.
3(a)	A gravitational field is a region of space in which a test mass experiences an